

Assignment 3: RAM and Storage Theory

Task 1: Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Table 1. Comparison of DDR3, DDR4, DDR5

Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Speed	800-2133 MT/s	1600-3200 MT/s	4800-8400+ MT/s
Voltage	1.5 V	1.2 V	1.1V
Bandwidth	Lower	Higher than DDR3	Highest among all three
Power Efficiency	Lowest	Better than DDR3	Best
Maximum Module Capacity	Up to 16 GB	Up to 64 GB	128 GB or more
Performance	Suitable for basic computing	Good for gaming, multitasking, and productivity	Excellent for high-end gaming, content creation, AI, and professional workloads
Compatibility	Works only with DDR3 motherboards	Works only with DDR4 motherboards	Works only with DDR5 motherboards
Typical Use Case	Older desktops and laptops, office work, web browsing	Modern desktops, laptops, gaming PCs, and workstations	Latest gaming PCs, high-performance desktops, advanced workstations, and servers
Cost	Lowest	Moderate	Highest

Task 2: Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully. Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.

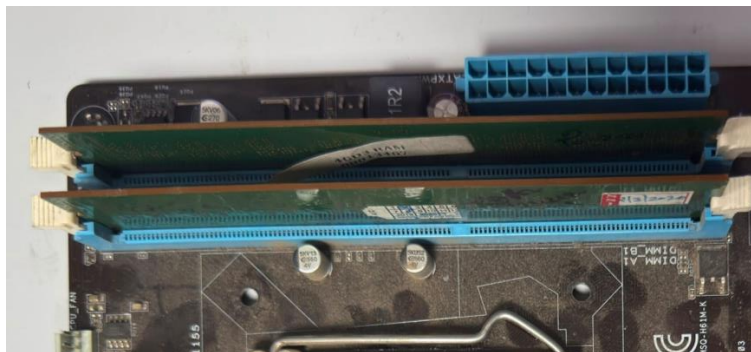


Fig. 1. Installed RAM Module

Task 3: List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Table 2. Comparison of HDD, SSD, NVMe

Feature	HDD	SSD	NVMe SSD
Speed	Slowest	Faster	Fastest
Technology	Uses spinning magnetic disks and a mechanical read/write head	Uses flash memory with no moving parts	Uses flash memory and connects through the PCIe interface using the NVMe protocol
Typical Use Case	Storing large files, backups, movies, and photos	Everyday computing, faster boot times, office work, and general gaming	High-performance gaming, video editing, software development, and professional workloads

Task 4: Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly. Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

<https://youtu.be/o9g4dn0qtJQ?si=wcQ6Xoq8gBcUtXlu>

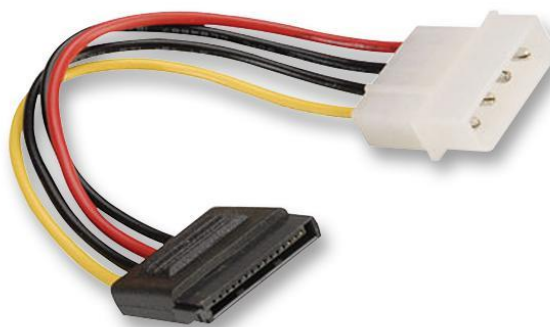


Fig 2: SATA Cable